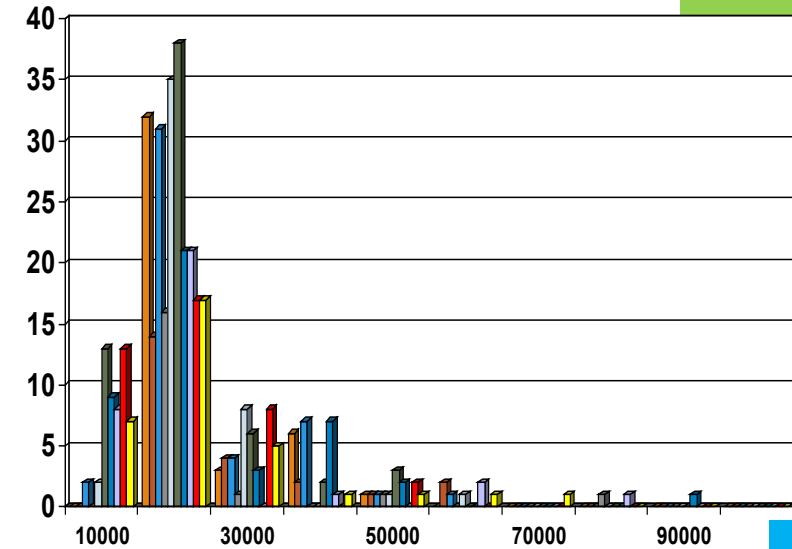


Histogram Analysis

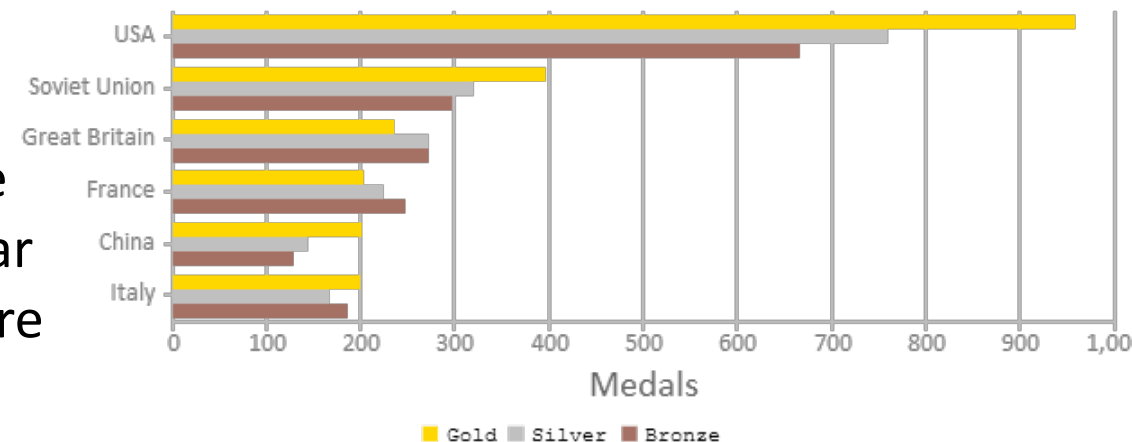
- ❑ Histogram: Graph display of tabulated frequencies, shown as bars
- ❑ Differences between histograms and bar charts
 - ❑ Histograms are used to show distributions of variables while bar charts are used to compare variables
 - ❑ Histograms plot binned quantitative data while bar charts plot categorical data
 - ❑ Bars can be reordered in bar charts but not in histograms
 - ❑ Differs from a bar chart in that it is the area of the bar that denotes the value, not the height as in bar charts, a crucial distinction when the categories are not of uniform width

Histogram

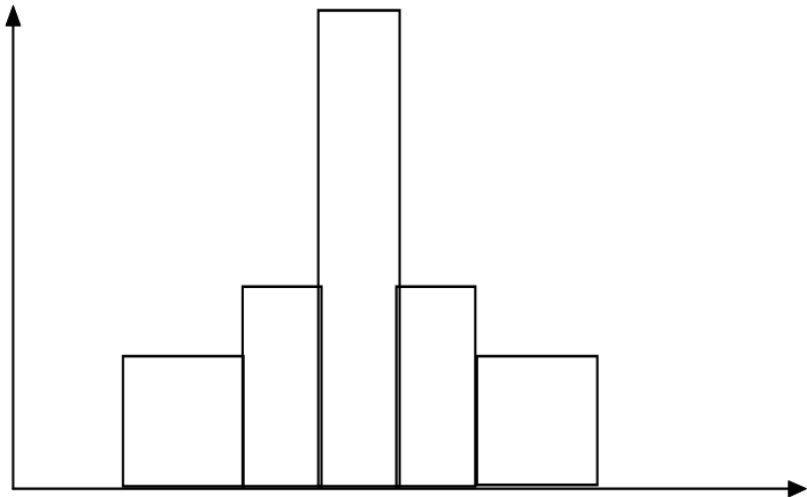
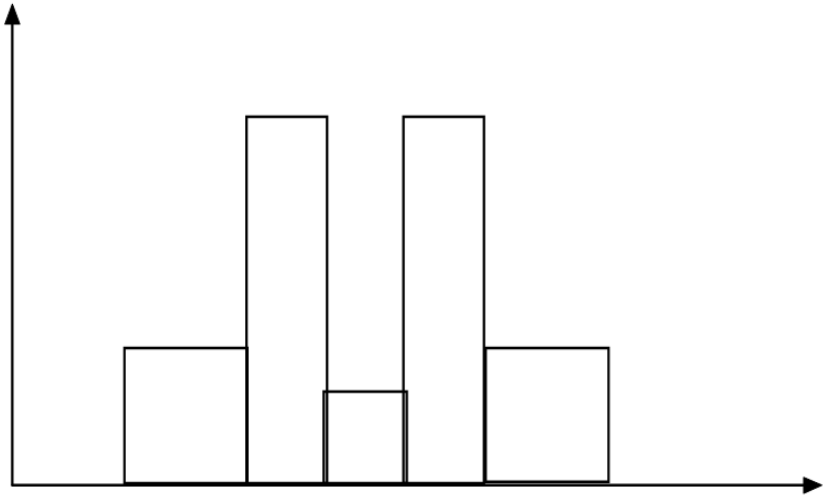


Bar chart

Olympic Medals of all Times (till 2012 Olympics)



Histograms Often Tell More than Boxplots



- The two histograms shown in the left may have the same boxplot representation
- The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions